

Values associated with the Distribution of the Average Range

		Subgroup Size (m)																			
		2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	
Number of Subgroups (g)	1	1.0 1.41421	2.0 1.91155	2.9 2.23887	3.8 2.48124	4.7 2.67253	5.5 2.82981	6.3 2.96288	7.0 3.07794	7.7 3.17905	8.3 3.26909	9.0 3.35016	9.6 3.42378	10.2 3.49116	10.8 3.55333	11.3 3.61071	11.9 3.66422	12.4 3.71424	12.9 3.76118	13.4 3.80537	
	2	1.9 1.27931	3.8 1.80538	5.7 2.15069	7.5 2.40484	9.2 2.60438	10.8 2.76779	12.3 2.90562	13.8 3.02446	15.1 3.12869	16.5 3.22134	17.8 3.30463	18.4 3.38017	19.0 3.44922	20.2 3.51287	22.4 3.57156	23.5 3.62625	24.5 3.67734	25.5 3.72524	26.5 3.77032	
	3	2.8 1.23105	5.7 1.76858	8.4 2.12049	11.1 2.37883	13.6 2.58127	16.0 2.74681	18.3 2.88628	20.5 3.00643	22.6 3.11173	24.6 3.20526	26.5 3.28931	28.4 3.36550	30.1 3.43512	31.9 3.49927	33.5 3.55842	35.1 3.61351	36.7 3.66495	38.2 3.71319	39.7 3.75857	
	4	3.7 1.20621	7.5 1.74989	11.2 2.10522	14.7 2.36571	18.1 2.56964	21.3 2.73626	24.4 2.87656	27.3 2.99737	30.1 3.10321	32.7 3.19720	35.3 3.28163	37.7 3.35815	40.1 3.42805	42.4 3.49246	44.6 3.55183	46.7 3.60712	48.8 3.65875	50.8 3.70715	52.8 3.75268	
	5	4.6 1.19105	9.3 1.73857	13.9 2.09601	18.4 2.35781	22.6 2.56263	26.6 2.72991	30.4 2.87071	34.0 2.99192	37.5 3.09808	40.8 3.19235	44.0 3.27701	47.1 3.35372	50.1 3.42381	52.9 3.48836	55.7 3.54787	58.4 3.60328	61.0 3.65502	63.5 3.70352	65.9 3.74914	
	6	5.5 1.18083	11.1 1.73099	16.7 2.08985	22.0 2.35253	27.0 2.55795	31.8 2.72567	36.4 2.86680	40.8 2.98829	45.0 3.09467	49.0 3.18911	52.8 3.27392	56.5 3.35077	60.1 3.42097	63.5 3.48563	66.8 3.54522	70.0 3.60072	73.1 3.65253	76.1 3.70109	79.1 3.74678	
	7	6.4 1.17348	12.9 1.72555	19.4 2.08543	25.6 2.34875	31.5 2.55460	37.1 2.72263	42.5 2.86401	47.6 2.98568	52.4 3.09222	57.1 3.18679	61.6 3.27172	65.9 3.34866	70.0 3.41894	74.0 3.48368	77.9 3.54333	81.6 3.59888	85.3 3.65075	88.8 3.69936	92.2 3.74509	
	8	7.2 1.16794	14.8 1.72147	22.1 2.08212	29.2 2.34591	36.0 2.55208	42.4 2.72036	48.5 2.86192	54.3 2.98373	59.9 3.09039	65.2 3.18506	70.3 3.27006	75.2 3.34708	80.0 3.41742	84.6 3.48221	89.0 3.54192	93.3 3.59751	97.4 3.64941	101.4 3.69806	105.3 3.74382	
	9	8.1 1.16361	16.6 1.71828	24.9 2.07953	32.9 2.34370	40.4 2.55013	47.7 2.71858	54.5 2.86028	61.1 2.98221	67.3 3.08896	73.3 3.18370	79.1 3.26878	84.6 3.34585	90.0 3.41624	95.1 3.48107	100.1 3.54081	104.9 3.59644	109.5 3.64838	114.1 3.69705	118.5 3.74284	
	10	9.0 1.16014	18.4 1.71573	27.6 2.07746	36.5 2.34192	44.9 2.54856	52.9 2.71717	60.6 2.85898	67.8 2.98100	74.8 3.08781	81.5 3.18262	87.9 3.26775	94.0 3.34486	99.9 3.41529	105.6 3.48016	111.2 3.53993	116.5 3.59559	121.7 3.64755	126.7 3.69625	131.6 3.74205	
	11	9.9 1.15729	20.2 1.71363	30.4 2.07577	40.1 2.34048	49.4 2.54728	58.2 2.71600	66.6 2.85791	74.6 2.98000	82.2 3.08688	89.6 3.18174	96.6 3.26690	103.4 3.34406	109.9 3.41452	116.2 3.47941	122.3 3.53921	128.1 3.59489	133.8 3.64687	139.4 3.69558	144.7 3.74141	
	12	10.7 1.15490	22.0 1.71189	33.1 2.07436	43.7 2.33927	53.8 2.54621	63.5 2.71504	72.6 2.85702	81.3 2.97917	89.7 3.08610	97.7 3.18100	105.4 3.26620	112.7 3.34339	119.9 3.41387	126.7 3.47879	133.3 3.53861	139.8 3.59430	146.0 3.64630	152.0 3.69503	157.9 3.74087	
	13	11.6 1.15289	23.8 1.71041	35.8 2.07316	47.3 2.33824	58.3 2.54530	68.7 2.71422	78.6 2.85627	88.1 2.97847	97.1 3.08544	105.8 3.18037	114.1 3.26561	122.1 3.34282	129.8 3.41333	137.3 3.47826	144.4 3.53810	151.4 3.59381	158.1 3.64582	164.7 3.69457	171.0 3.74041	
	14	12.5 1.15115	25.7 1.70914	38.6 2.07213	51.0 2.33737	62.8 2.54452	74.0 2.71351	84.7 2.85562	94.9 2.97787	104.6 3.08487	113.9 3.17984	122.9 3.26510	131.5 3.34233	139.8 3.41286	147.8 3.47781	155.5 3.53766	163.0 3.59339	170.3 3.64541	177.3 3.69417	184.2 3.74002	
	15	13.4 1.14965	27.5 1.70804	41.3 2.07125	54.6 2.33661	67.2 2.54385	79.3 2.71290	90.7 2.85506	101.6 2.97735	112.1 3.08438	122.1 3.17938	131.7 3.26465	140.9 3.34191	149.8 3.41245	158.3 3.47742	166.6 3.53728	174.6 3.59302	182.4 3.64505	190.0 3.69382	197.3 3.73969	
	16	14.3 1.14833	29.3 1.70708	44.1 2.07047	58.2 2.33594	71.7 2.54326	84.5 2.71237	96.7 2.85457	108.4 2.97689	119.5 3.08395	130.2 3.17897	140.4 3.26427	150.2 3.34154	159.7 3.41210	168.9 3.47707	177.7 3.53695	186.3 3.59270	194.6 3.64474	202.6 3.69351	210.4 3.73939	
	17	15.1 1.14717	31.1 1.70623	46.8 2.06978	61.8 2.33535	76.2 2.54274	89.8 2.71190	102.8 2.85413	115.1 2.97649	127.0 3.08358	138.3 3.17861	149.2 3.26393	159.6 3.34121	169.7 3.41178	179.4 3.47677	188.8 3.53666	197.9 3.59242	206.7 3.64447	215.2 3.69325	223.6 3.73913	
	18	16.0 1.14613	32.9 1.70547	49.5 2.06917	65.5 2.33483	80.6 2.54228	95.1 2.71148	108.8 2.85375	121.9 2.97613	134.4 3.08324	146.4 3.17829	157.9 3.26362	169.0 3.34092	179.7 3.41150	189.6 3.47650	199.9 3.53640	209.5 3.59216	218.8 3.64422	227.9 3.69301	236.7 3.73890	
	19	16.9 1.14520	34.7 1.70480	52.3 2.06862	69.1 2.33436	85.1 2.54187	100.3 2.71111	114.8 2.85341	128.7 2.97581	141.9 3.08294	154.5 3.17801	166.7 3.26335	178.4 3.34066	189.6 3.41125	199.5 3.47626	210.5 3.53617	221.1 3.59194	231.0 3.64400	240.5 3.69280	249.8 3.73869	
	20	17.8 1.14437	36.5 1.70419	55.0 2.06813	72.7 2.33394	89.6 2.54149	105.6 2.71077	120.9 2.85310	135.4 2.97552	149.3 3.08267	162.7 3.17775	175.5 3.26311	187.8 3.34042	199.6 3.41103	211.0 3.47605	222.1 3.53596	232.8 3.59174	243.1 3.64380	253.2 3.69260	263.0 3.73850	
d ₂		1.12838	1.69257	2.05875	2.32593	2.53441	2.70436	2.8472	2.97003	3.07751	3.17287	3.25846	3.33598	3.40676	3.47193	3.53198	3.58788	3.64006	3.68896	3.735	
cd		0.876	1.815	2.7378	3.623	4.4658	5.2673	6.0305	6.7582	7.4539	8.1207	8.7602	9.3751	9.9679	10.5396	11.0913	11.6259	12.144	12.6468	13.1362	

Table entries: 1st line of each cell is the degrees of freedom (v) and the 2nd line of each cell is d_2 ; d_2 is the infinity value of d_2 ; Additional values of v can be built up from the constant difference cd.

Note: The notation used in this table follows that of Acheson Duncan *Quality Control and Industrial Statistics*, 5th edition, McGraw-Hill, 1986:

$\sqrt{(\bar{R}/d_2)}/\sigma^{1/2}$ is distributed approximately as a χ^2 distribution with v of freedom where $\bar{R}$ is the average range of g subgroups of size m

Source : "Measurement Systems Analysis", Reference Manual, © DaimlerChrysler Corporation, Ford Motor Company, General Motors Corporation, 2002.